

Subject Description Form

Subject Code	EIE4102
Subject Title	IP Networks
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	EIE3333 Data and Computer Communications
Objectives	1. Give a practical treatment on the design, implementation, and management of IP networks. 2. Introduce the variety of facilities, technologies, and communication systems to meet future needs of network services. 3. Evaluate critically the performance of existing and emerging global communication networking technologies.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Describe the operational and functional attributes of different components of IP networks. 2. Evaluate critically the design, implementation, and performance of IP networks with regard to different criteria. <u>Category B: Attributes for all-roundedness</u> 3. Think and evaluate critically. 4. Take up new technology for life-long learning.
Subject Synopsis/ Indicative Syllabus	1. <u>Basic Protocol Functions</u> IP address, IP datagram structure, basic IP operations, delivery and forwarding IP packets 2. <u>Protocols in TCP/IP</u> ARP, RARP, ICMP, IGMP, UDP, TCP 3. <u>Routing Protocols</u> RIP, OSPF, BGP, Multicast Routing 4. <u>Applications Over TCP/IP</u> DNS, TELNET, FTP, Email, HTTP 5. <u>Other Issues About IP</u> IP over ATM, Mobile IP, Multimedia, Voice over IP, SIP, H.323, IPv6, IPSec Laboratory Experiments: 1. Voice over IP Experiment 2. IP Security

Teaching/Learning Methodology	Teaching and Learning Method	Intended Subject Learning Outcome	Remarks				
	Lectures	1, 2	Fundamental principles and key concepts of the subject are delivered to students.				
	Tutorials	1, 2, 3, 4	Supplementary to lectures. Students will be able to clarify concepts and to have a deeper understanding of the lecture material; Problems and application examples are given and discussed.				
	Laboratory sessions	2,3,4	Students will conduct practical exercises to reinforce concepts and techniques learned.				
Assessment Methods in Alignment with Intended Subject Learning Outcomes	Specific Assessment Methods/Tasks		% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)			
				1	2	3	4
	1. Continuous Assessment (total: 50%)						
	• Assignments		10%	✓	✓	✓	
	• Laboratory reports		10%		✓	✓	✓
	• Mid-Term Test		15%	✓	✓	✓	✓
	• End-of-Term Test		15%	✓	✓	✓	✓
	2. Examination		50%	✓	✓	✓	✓
	Total		100%				
Student Study Effort Expected	Class contact (time-tabled):						
	• Lecture			24 Hours			
	• Tutorial/Laboratory/Practice Classes			15 Hours			
	Other student study effort:						
	• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes/examination			36 Hours			
	• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing			30 Hours			
	Total student study effort:			105 Hours			

Reading List and References	1. Behrouz A. Forouzan, <i>TCP/IP Protocol Suite</i>, 3rd ed., McGraw-Hill, 2006. 2. Howser, Gerry, <i>Computer Networks and the Internet: A Hands-On Approach</i>, Cham: Springer International Publishing AG, 2019.
------------------------------------	---

July 2025